

Raw SERS data



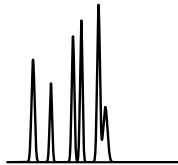
SERS map

one spectrum
per pixel



NNLS gate

background
dropped

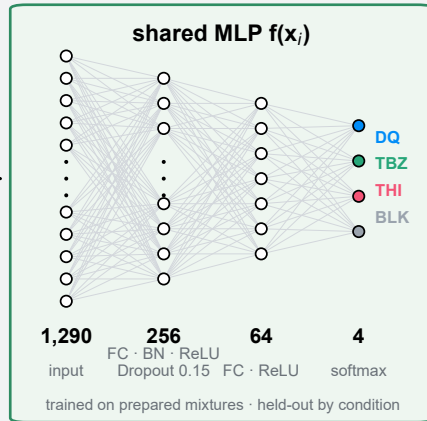


pixel spectra x_i

1,290 pts · 500–2500 cm^{-1}

Compositional unmixing

output



distribution mapping

DQ TBZ THI



gated pixels · composition

map composition (%)

THI ~34

TBZ ~32

DQ ~28

BLK ~6 (removed)



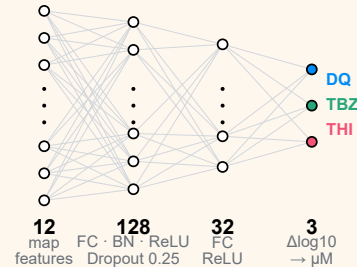
absent component $\rightarrow 0$ (absence gate, $P < 0.2$)

Concentration

output

concentration net $g(\text{map})$

12 map features: $\log_{10} C_{\text{cal}} \cdot 3 \cdot \text{composition} \cdot 3 \cdot \log_{10} \text{band} \cdot 3 \cdot \log\text{-total p10/50/90}$



domain check: nearest training map z-distance ≤ 3 , else refused

declared total (optional): $\hat{C}_i = \bar{p}_i \cdot C_{\text{total}}$